Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/24/2025
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Project Title:	Effect of GCR Exposure on the Functional Status of the DNA Repair Machinery and Its Contribution to Cancer Risk		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
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Zip Code:	30322-4218	**Congressional District:**	5
Comments:			
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Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
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Key Personnel Changes/Previous PI:	May 2019 report: None ; May 2020 report: None. May 2021 report: none.		
COI Name (Institution):	Yu, David M.D., Ph.D. (Emory University)		
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Task Description:	Current predictions of the risk of Radiation Exposure Induced Death have large uncertainties. In part, this is because of poor understanding of factors that influence risk in the low dose range corresponding to anticipated human exposure. Modeling suggests that the dose response may be strongly nonlinear, with an increase in risk at low doses attributable to Non-Targeted Effects (NTEs) in cells not directly traversed by a radiation track. A model incorporating NTEs, fitted to mouse Harderian gland carcinogenesis data, predicts that Mars mission cancer risk may be two-fold greater than that attributable to targeted effects alone. These findings highlight the need for a better understanding of the mechanisms underlying the high Relative Biological Effectiveness (RBE) of Galactic Cosmic Radiation (GCR) for carcinogenesis, including its ability to elicit NTEs. The principal hypothesis for this proposal is that exposure to GCR, and particularly its high charge and energy (HZE) particle components, is associated with chronic DNA damage signaling leading to decline in the functional status of the DNA repair machinery, and that this is a key contributor to GCR carcinogenesis. Prior studies establish the scientific premise for this hypothesis and identify potential underlying mechanisms, one of which is depletion of cellular NAD+ pools as a consequence of persistent DNA damage, resulting in reduced activity of important NAD+-dependent enzymes. These include the DNA damage sensing and repair factor, PARP1, and protein deacylases known as sirtuins. Experiments will investigate the role of the NAD+/PARP1/sirtuin axis in experiments using both human bronchial epithelial cells and wild-type C57Bl/6 mice. The work will focus on lung cancer risk as an endpoint. Specific Aims are: Aim 1. To investigate the scope and mechanistic basis for a GCR exposure-related decline in the functional status of the repair machinery using a bronchial epithelial cell model. (Removed from Study) Aim 2. To test the hypothesis that GCR exposure leads to a chronic DNA damage response in mouse lung and whether this can be abated with a dietary countermeasure. We will investigate gender, dose, and radiation quality effects. Aim 3. To determine whether radiogenic tumors display a mutational signature indicative of a decline in the functional status of the DNA repair machinery as reflected in their mutational signature, thus establishing a functional link to tumorigenesis. Anticipated deliverables include research results to improve evaluation of sex dependence of radiogenic lung cancer; to allow for calculation of RBEs including mixed field RBEs for males and females, particularly at low doses and in wild-type backgrounds where direct measurements of RBE for radiation-induced cancers is infeasible or subject to large experimental uncertainties; and to perform an initial evaluation of a potential countermeasure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The nicotinamide riboside countermeasure may have anti-aging properties that would make it useful outside the context of galactic cosmic ray exposure. Results may bear on this question.
Task Progress:	Study design. These were ground-based studies in which mice (strain C57/Bl6, age 24 weeks) were irradiated at the NASA Space Radiation Laboratory using 28-Si ions, a 6-ion simplified galactic cosmic ray simulation (sGCR), or γ-ray reference radiation. Non-irradiated mice were handled in parallel. Some mice received a potential countermeasure, 12 mM nicotinamide riboside in drinking water, starting 48 h before irradiation and continuing throughout their life. A total of 2,040 mice were used in the study. Some mice were sacrificed after 1 or 6 months for tissue and biomarker analysis. Others were maintained for 18 months for the tumorigenesis endpoint. The experiment was designed to allow determination of radiation effects ratios at a fixed dose (0.75 Gy), determination of a countermeasure effect for 28-Si only, and determination of a dose response in select instances. Cost considerations dictated a sparse design, where only some groups could be included for the 18-month tumor endpoint. Chronic DNA damage signaling is reflected in protein PARylation levels. Single-and double-strand DNA breaks, as well as replication stress, activate the PARP1 or PARP2 enzymes, which catalyze a unique post-translational modification involving the addition of highly-charged, branched PAR chains. Elevated levels of PARylated proteins are thus an indicator of chronic DNA damage signaling. One of the principal substrates for PARP1 is the enzyme itself. This auto-modification, which occurs when the enzyme is bound at sites of DNA damage, can be detected based on a mobility shift in SDS-PAGE and reactivity with anti-PAR antibody. An immunoblotting assay for PARylated PARP-1 was established, validated using recombinant PARP1, then used to analyze PARylation of endogenous PARP1 in extracts of flash-frozen mouse lung tissue. In most instances, 5 mice were analyzed for each radiation type/timepoint/dose/treatment/sex combination. At 1 mo post irradiation, PARylated PARP1 was significantly induced by 28-Si (0.75 Gy) or sGCR (0.75 Gy), but not by a lower dose of 28-Si, a higher dose of sGCR, or any dose of γ-rays. Males and females responded similarly. Results with nicotinamide riboside, at a maximum tolerated dose, remain proprietary. Analysis of samples collected at 6 mo post-irradiation showed uniformly higher levels of PARylation, perhaps reflecting more advanced age of the mice, with no easily understood relationship to treatment. Separate experiments were performed using anti-PAR immunohistochemistry to analyze formalin-fixed contralateral lungs of the same mice. Positive control (acutely irradiated) mice showed strong staining, as did mice irradiated with 28-Si and analyzed after 1 mo. Results with other treatment groups were consistent with immunoblotting, although results proved challenging to quantify. Exploratory studies of other potential biomarkers. Staining was performed for γ-H2AX foci in lungs of mice irradiated with 0.75 Gy of 28-Si ions or 1.5 Gy of γ-rays and collected at 1 mo post-irradiation. No foci were detected. In a later experiment, γ-H2AX staining was performed with various specimens collected at 6 mo post-irradiation. Aside from a few fields with inflammatory infiltrates, only scattered, isolated γ-H2AX-positive nuclei were present. Results suggest that γ-H2AX is not a useful marker of genomic stress in lung tissue. Exploratory staining for CD68 (a marker of macrophage infiltration), phospho-XRCC1 (a marker of single-strand DNA break repair), and 4-hydroxynonenal (a marker of lipid peroxidation) showed no consistent differences between irradiated and non-irradiated mice. These potential markers were not pursued further. Lung tumor incidence. As a modest expansion of the original aims, the size of some experimental groups was set at 200 mice, and these were held for 18 mo post-irradiation to examine tumor incidence. An average of 156 evaluable specimens per group were obtained, reflecting that some mice died prematurely or were removed for other reasons. Mortality did not differ significantly between irradiated and non-irradiated groups. Common causes of premature mortality included non-healing skin ulcers, injuries from fighting, or tumors (mostly of the liver). Some mice also

	succumbed to unknown causes. Lung tumor incidence. Among long-term survivors, the baseline incidence of lung tumors for non-irradiated male mice was 22.1%. This incidence was much higher than in an earlier study, performed under somewhat different conditions, where no tumors were reported in non-irradiated mice (doi: 10.1667/RR13884.1). The incidence of lung tumors in irradiated male mice ranged from 21.0% to 27.7%. There was no significant effect of radiation dose, radiation quality, or countermeasure compared to control. Overall, the high incidence of tumors in control mice confounded the ability to measure a significant radiation effect. As in the prior study, almost all (90%) of the lung tumors were small adenomas of uncertain significance. The baseline incidence among female mice was 4.0% following 0.75 Gy sGCR and 5.1% following 0.75 Gy γ-ray irradiation. This difference was not statistically significant. Tumorigenesis studies in female mice were performed with the sole objective of comparing relative incidence in groups subjected to sGCR and γ-rays, and therefore, there was no non-irradiated control group. Incidental finding of inflammation. Some lung specimens showed incidental findings of BALT hyperplasia, eosinophilic crystalline pneumonia, or alveolar histiocytosis. Three groups showed elevated incidence relative to controls: 0.75 28Si-irradiated mice, 1.5 Gy sGCR-irradiated mice, and 1.5 Gy γ-rays. Lower doses of sGCR or γ-rays did not elicit the effect. Results with countermeasure remain proprietary. Although these are incidental findings and not a planned study endpoint, they do suggest a long-term inflammatory effect that is more severe with heavy ions than with the mixed radiation or γ-rays, and with some mitigation by countermeasure. Summary and conclusions. Studies of protein PARylation in the lungs of irradiated mice suggest persistent genomic stress in 28-Si and sGCR irradiated groups, with evidence of protection by countermeasure. Further study will be needed to understand a nonlinear dose response and the mechanism of action for the countermeasure. Results with other candidate biomarkers were negative. Spontaneous incidence of small lung adenomas was higher than expected based on an earlier report, possibly reflecting the use of older mice in the present study (which was done to better simulate risk to middle-aged astronauts). Baseline tumor incidence among male mice was much higher than among females. Irradiation did not significantly increase incidence over the high baseline. There was an incidental finding of inflammatory changes in some irradiated groups. Results do not support a hypothesis, based on the earlier study, that the C57BL/6 genetically unmodified (non-sensitized) mouse model can be used to investigate the risk of space-radiation exposure-induced lung tumorigenesis.
Bibliography Type:	Description: (Last Updated: 08/06/2025)
Articles in Peer-reviewed Journals	Hertel NE, Biegalski SR, Nelson VI, Nelson WA, Mukhopadhyay S, Su Z, Chan AM, Kesarwala AH, Dynan WS. "Compact portable sources of high-LET radiation: Validation and potential application for galactic cosmic radiation countermeasure discovery." Life Sci Space Res. 2022 Nov;35:163-69. https://doi.org/10.1016/j.lssr.2022.10.002 ; PMID: 36336362 , Nov-2022